

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Name	Rising Waters

1. PROJECT OVERVIEW	
Section	Description
Purpose	To develop an AI-powered Raising Water Monitoring and Flood Alert System that collects real-time water level data from various sources, analyzes the data, and provides early flood warnings to help minimize loss of life and property.
Scope	The project covers real-time water level monitoring of rivers, lakes and reservoirs using IoT sensors, weather data integration, AI/ML-based flood risk prediction, alert generation and a user-friendly web & mobile dashboard for visualization and notifications.
Target Users	Residents in flood-prone areas, local authorities, disaster management teams, government agencies, and NGOs.
Goal	To build an accurate, reliable and real-time flood monitoring and alert system that helps communities prepare in advance and respond quickly to flood situations by providing timely alerts and risk assessment updates.

2. PROBLEM STATEMENT	
Section	Description
Context	In many regions, rising water levels due to heavy rainfall and poor water management often lead to flooding, which causes severe damage to life, property, agriculture and infrastructure.
Issue	There is a lack of real-time water level monitoring and accurate flood prediction systems. Traditional methods rely on manual observation and delayed information.
Key Impact	- Delayed flood warnings lead to loss of life and property. - Communities are not well-prepared for sudden floods. - Lack of awareness and timely information increases disaster risks.

3. PROPOSED SOLUTION

Section	Description
Solution	A Raising Water Monitoring and Flood Alert System that uses IoT sensors, weather data and AI/ML models to predict flood risk and send instant alerts to users through a web dashboard and mobile notifications.
Key Features	- Real-time water level monitoring using IoT sensors. - Weather data integration and rainfall analysis. - AI/ML based flood prediction and risk assessment. - Instant alerts via SMS, Email and In-app notifications. - Dashboard for live monitoring, maps and reports.
Expected Outcome	- Early flood warnings and risk prediction. - Increased preparedness and quick response to floods. - Reduced loss of life, property and resources.

4. RESOURCE REQUIREMENTS

Resource Type	Description	Specification / Allocation
Hardware	System for development and testing	Processor: Intel Core i3 or above, RAM: Minimum 4 GB, Storage: Minimum 10 GB free space
Internet	Internet Connectivity	Required for downloading packages, datasets and deployment
Operating System	Platform for development	Windows / Linux / macOS
Framework	Web framework	Flask (Python)
Libraries	Core libraries for ML & data handling	scikit-learn, pandas, numpy, SQLAlchemy, joblib, requests
Development Environment	IDE / Environment	Visual Studio Code, Jupyter Notebook / PyCharm
Data Source, Size, Format	Dataset for model training	Historical water level data, rainfall data (Govt/Public datasets), CSV format